

UNGGI LEE

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EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) Daegu, S.Korea
M.S. in Computer Science Feb. 2026 - Feb. 2028 (Expected)

- Real-Time Computing Lab (RTCL)

Daegu Gyeongbuk Institute of Science and Technology (DGIST) Daegu, S.Korea
B.S. in Electrical Engineering and Computer Science Feb. 2020 - Feb. 2026

- Total GPA of 3.82 / 4.30

RESEARCH INTERESTS

Real-time systems; DNN inference offloading between edge devices and servers; image-adaptive compression under bandwidth variation; the trade-off between data fidelity, transmission cost, and end-to-end deadlines.

RESEARCH EXPERIENCE

Real-Time Computing Lab (RTCL), DGIST Daegu, S.Korea
Graduate Research Assistant Feb. 2026 - Present

- Real-time systems, with a focus on DNN inference offloading between edge devices and servers under latency constraints.
- Image-adaptive JPEG compression: per-image quality factor and quantization table optimization, so that offloaded inference holds up under variable network bandwidth.

PUBLICATIONS

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1. Minseok Oh, **Unggi Lee**, Hyunmin Kim, Dooseok Lee. "WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation." *2025 16th International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 1753-1758, October 2025. doi:10.1109/ICTC66702.2025.11388251.

PROJECT EXPERIENCE

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation Daegu, S.Korea
Reinforcement Learning course project, published at ICTC 2025 2025

- A model-free PPO framework that assembles top-bottom outfit pairs from a natural-language style prompt, rewarded by CLIP similarity fused with a learned proxy for human preference.

Noise Map-Driven Dynamic Frame Processing for Improved Object Detection in Low-Light Conditions Daegu, S.Korea
Undergraduate Group Research Program (UGRP), Team of Four 2024

- Placed an enhancement front-end before YOLO — Zero-DCE for low-light enhancement, CBDNet for noise-map estimation, and MFDNet for denoising — integrated directly into the detector's data loader.
- Held the pipeline to real time by exploiting temporal redundancy between frames, processing one frame in five at 20 FPS.
- Improved mAP by roughly 1% over the unenhanced baseline on BDD100k.

INTERNATIONAL EXPERIENCE

University of California, Los Angeles (UCLA)

Summer Session

Los Angeles, CA, USA

Jun. - Aug. 2023

SKILLS

Programming: C, C++, Python

Deep Learning: PyTorch

Computer Vision: OpenCV

Tools: Docker, Linux, Git

CONFERENCE PRESENTATIONS

"WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation." *2025 16th International Conference on Information and Communication Technology Convergence (ICTC)*, October 2025.